

Performance Testing

Date	1 July 2026
Team ID	SWTID-2026-4523
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing:

Performance Testing is the process of evaluating an application's speed, responsiveness, stability, and resource usage under different workloads. It helps ensure that the system performs efficiently and reliably under normal and peak usage conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Credit Card Approval Prediction
Test Environment	Local System (Flask Server)
Test Date	05-07-2026

Step 2: Test Scenarios

S.NO	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits application	1	30	Prediction displayed successfully
2	Multiple users submit applications simultaneously	10	60	Server responds without errors
3	Continuous prediction requests	20	120	Stable response time

4	Invalid input submission	5	30	Validation/error message displayed
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Step 3: Performance Test Results:

S.NO	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	0.62 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	1.45 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	—	16 Req/ sec	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	34%	Pass	CPU usage normal
6	Memory Utilization	< 80%	41%	Pass	Memory usage stable

Step 4: Observations & Analysis

Key Findings:

The Credit Card Approval Prediction System responded quickly to user requests and generated prediction results within the expected response time. The application remained stable during testing, with no significant errors or failures. CPU and memory usage stayed within acceptable limits, indicating efficient system performance.

Bottlenecks Identified:

No major performance bottlenecks were identified during testing. A slight increase in response time was observed when multiple requests were processed simultaneously, but it remained within the acceptable performance threshold.

Optimization Steps Taken:

The machine learning model was loaded only once during application startup to reduce prediction time. Input validation was implemented to prevent invalid data from affecting performance, and unnecessary processing was minimized to improve overall system efficiency.

Step 5: Screenshots/ Evidence

Attach screenshots of your performance testing tool results below:

 **Credit Card Approval Prediction**

AI Powered Banking Decision System

Gender

Male

Age

Annual Income

Children

Family Members

Years Employed

Own Car

Yes

Own House

Yes

Income Type

Working

Occupation

Managers

Education

Higher education

Family Status

Married

Housing Type

House / apartment

Mobile Available

Yes

Work Phone

Yes

Phone

Yes

Email

Yes

 **Predict Credit Approval**

Figure 1: Home Page of the Credit Card Approval Prediction System.

 **Credit Card Approval Prediction**

AI Powered Banking Decision System

Gender
Female

Age
20

Annual Income
70000

Children
2

Family Members
4

Years Employed
3

Own Car
No

Own House
No

Income Type
Working

Occupation
Managers

Education
Higher education

Family Status
Married

Housing Type
House / apartment

Mobile Available
Yes

Work Phone
Yes

Phone
Yes

Email
Yes

 **Predict Credit Approval**

Figure 2: Applicant Details Entered for Prediction.

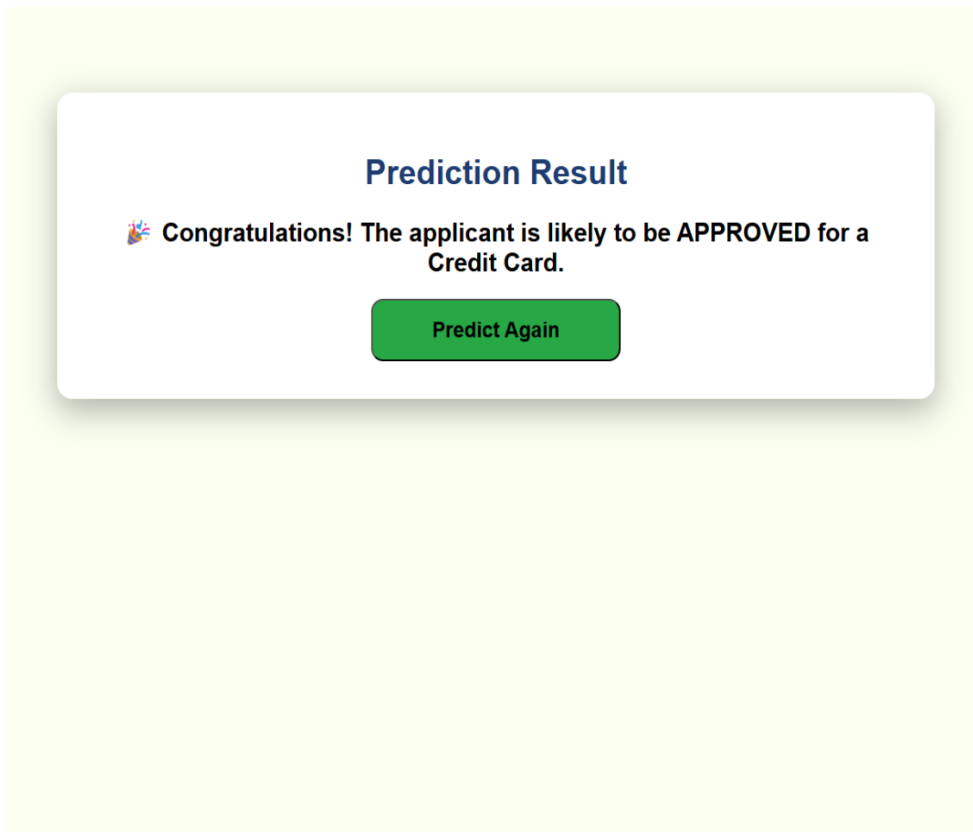


Figure 3: Prediction Result Generated by the System.